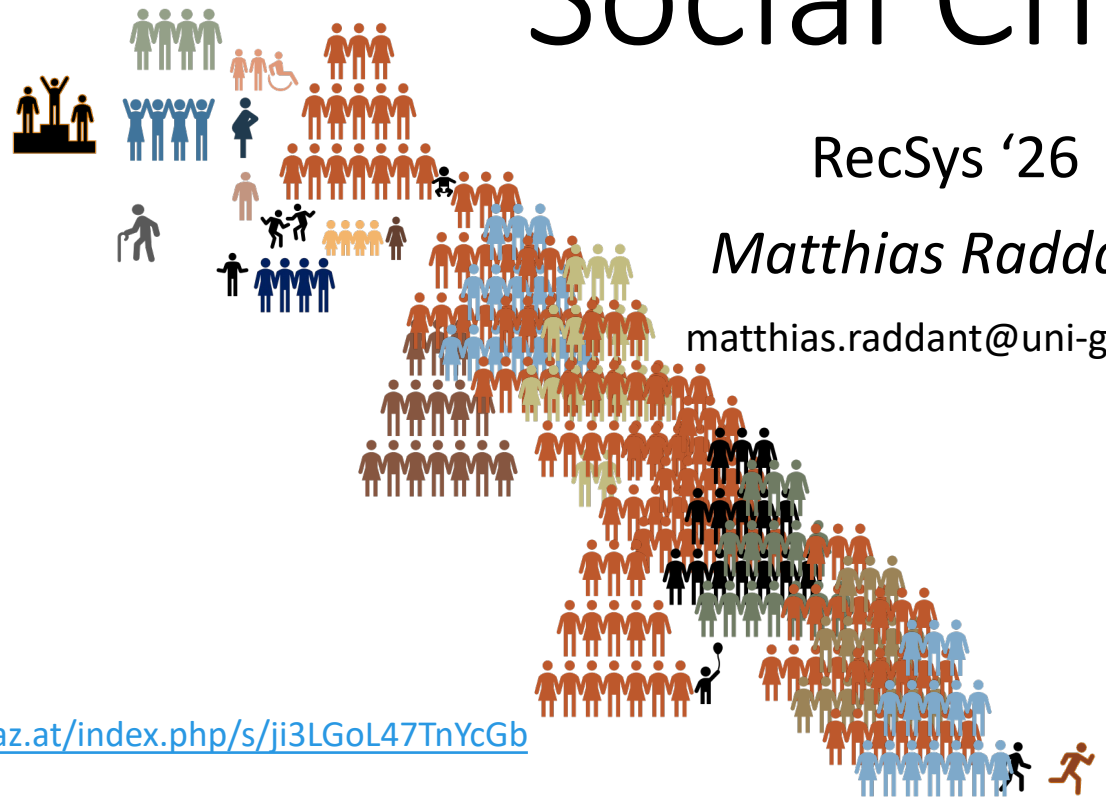
(Computational) Social Choice





RecSys ‘26

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video here:
https://cloud.tugraz.at/index.php/s/ji3LGoL47TnYcGb

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What is a Social Choice Problem?

- We choose collectively among options/outcomes with, e.g., with respect to
  - voting
  - allocation of resources, goods or services
  - matching, allocation of tasks
- Agents have **preferences** (opinions) about options/outcomes.
- The **social planer** knows the preferences (or they are revealed).
- Some social choice function is used to aggregate preferences and to choose an outcome.
- Relevance: decisions in shared environments/projects, elections, public goods, ratings, collaborative filtering

Views on Social Choice

- The foundations of social choice theory were developed at the intersection of mathematics, economics, political sciences and to some extend also sociology and law.

  -> When looking into the literature, be prepared to find work that emphasizes very different aspects of SC.

- Computational Social Choice is (only) around for about 20 years.
- In this lecture we will cover some SC basics and hint at some starting points for CSC.
- For more details see

Handbook of Computational Social Choice (2016), F. Brandt, V. Conitzer, U. Endriss, J. Lang, A.D. Procaccia, Cambridge UP.

RecSys vs. Social Choice

- Recommendations to
  - individuals
  - groups
- Based on
  - peers
  - content
  - knowledge
- Deal with biases of users/data
- Provide explaination
- Possibly collaborative interaction for group decission making

- Selecting outcomes for all
- Based on
  - preferences
  - social welfare
  - choice function
- Allocating/matching fixed resources
- Design mechanisms that do not provide incentives to “cheat” to manipulate rankings
- Interactions mostly in a strategic sense

Agenda

1. Voting
2. Allocation of resources
3. Matching
4. Social Interaction



from the Handbook of CSC

1. A classic example on voting

Assume 3 candidates and a population of 11 voters.

All voters rank all candidates.

Rankings of candidates			
4x	3x	2x	2x
Emanuel	Olaf	Olaf	Evika
Olaf	Evika	Emanuel	Emanuel
Evika	Emanuel	Evika	Olaf

ranking

$\Sigma = 11\ voters$

Who should be elected?

Plurality rule

Who is ranked on top most often? Olaf

Ranking of candidates			
4x	3x	2x	2x
Emanuel	**Olaf**	**Olaf**	Evika
Olaf	Evika	Emanuel	Emanuel
Evika	Emanuel	Evika	Olaf

Problem: No absolute majority, no evaluation of rank 2,3, …

Borda rule

Each candidate receives as many points from a voter as candidates are ranked below her.

Ranking of candidates			
4x	3x	2x	2x
Emanuel	Olaf	Olaf	Evika
Olaf	Evika	Emanuel	Emanuel
Evika	Emanuel	Evika	Olaf

Olaf receives: 4x1+3x2+2x2 and wins.

However: 6 voters prefer Emanuel over Olaf, 6 voters prefer Emanuel over Evika

Generalization: Arrow's theorem

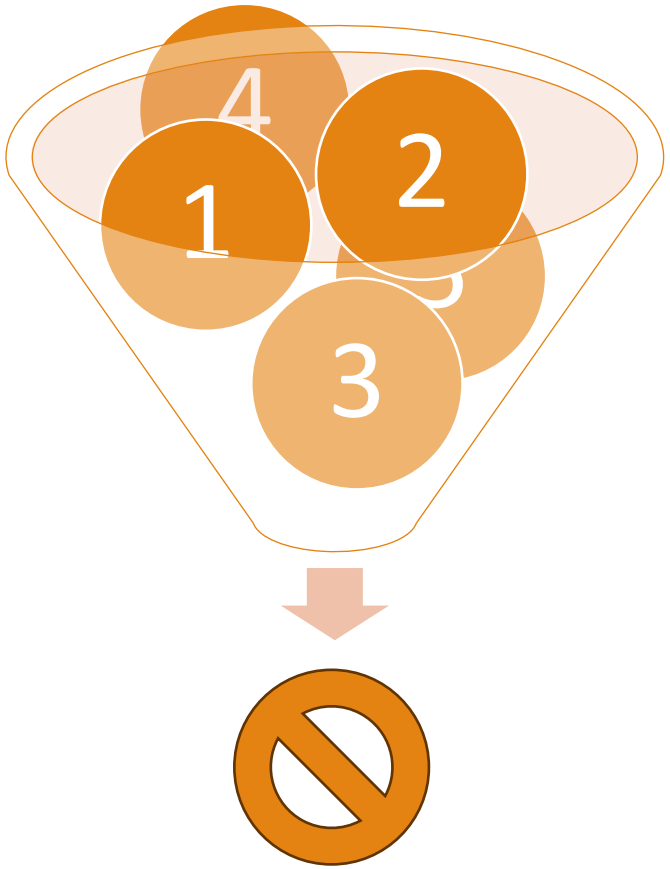
- Assume 3 agents and 3 options. All agents have preferences about the ordering of these options, e.g.,
  - 1. **A B C**
  - 2. **B C A**
  - 3. **C A B**
- What would be a socially preferred ordering of **A**,**B** and **C**?
- Problem: **A** is preferred to **B** two times, **B** is preferred over **C** two times, **C** is preferred over **A** two times
- Question: Is there a procedure (a social welfare function) that produces a ranking and that fulfils certain conditions on fairness?

The **5 conditions** (simplyfied)

1. **Unrestricted domain**: Social welfare/choice function (SWF) defined for all combinations of individual preferences
2. **Social ordering**: SWF produces an ordering for all possible combinations of preferences
3. **Weak Pareto**: If everyone prefers a certain option over another one, then the SWF also has to prefer this option
4. **Non-dictatorship**: Social preferences are not dependent on the ordering that one single agent has
5. Independence of **Irrelevant Alternatives**: SWF's ordering of 2 alternatives must be independent of any additional option

See also https://plato.stanford.edu/entries/arrows-theorem/

Morreau, Michael, "Arrow's Theorem", *The Stanford Encyclopedia of Philosophy* (Winter 2019 Edition), Edward N. Zalta (ed.).



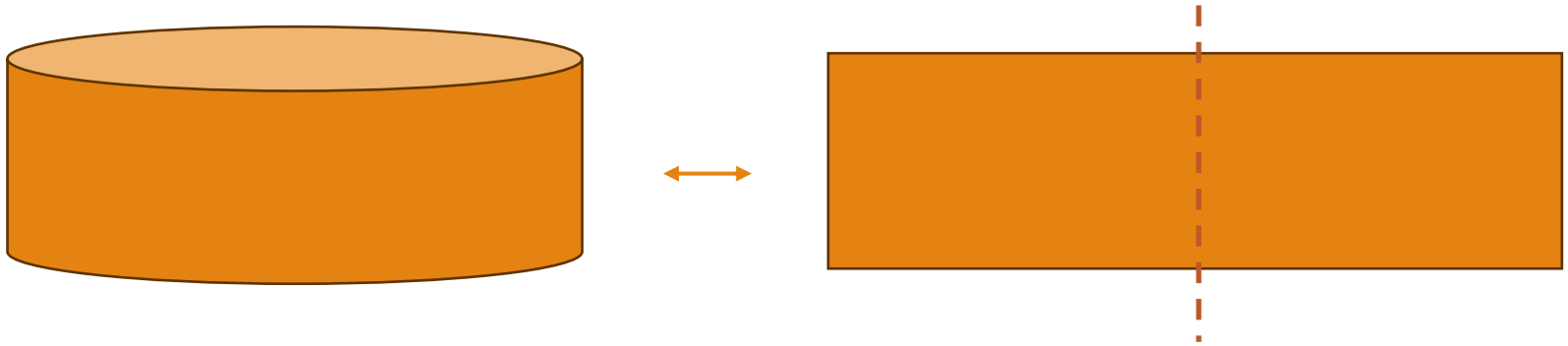
Arrow: This is not possible!

(Hence the “impossibility” theorem)

2. Fair Allocation: Cake Cutting

Assume a set of ***N*** agents and a heterogenous divisible good, in particular a “cake” in the [*0,1*] interval. Agents have a value function $V_i(I)$ for intervals of that cake.

We have to devide the entire cake such that each agent receives a fair share of the cake.



Criteria for fairness

- **Proportionality**: Each agent receives a piece that she values at

$$V_i(A_i) \geq 1/n$$

- **Envy-freeness**: Nobody else gets a piece with a higher value,

$$V_i(A_i) \geq V_i(A_j)$$

- **Equitability**: Everybody values his/her own piece equally,

$$V_i(A_i) = V_j(A_j)$$

An easy start: N=2

Cut and choose

- Agent 1 cuts two pieces that he values as equal
- Agent 2 chooses a piece, agent 1 receives the remaining piece
- Agent 1 has an incentive to cut two equal pieces. Even if the preferences of the agent differs, they will both receive a piece valued at least 1/n.

N=3, Selfridge-Conway algorithm

- Agent 1 devides cake in three pieces with $V_1(X_1)=V_1(X_2)=V_1(X_3)$
- Agent 2 removes a piece from the most valuable (for her) piece to equalize with the second most valuable one, creating a "cake 2".

Assume $V_2(X_1) > V_2(X_2) \geq V_2(X_3)$

Hence, we trim $\setminus X$ from piece X_1 and we are left with the pieces,

$X_1 \setminus X$, X_2 and X_3 that form cake 1.

Cutting cake 1

- Agent 3 chooses one piece from cake 1
- If 3 has selected the trimmed piece -> agent 2 chooses between the remaining pieces
- If 3 has selected the non-trimmed piece -> agent 2 receives the trimmed piece
- Name the agent $\in\{2,3\}$ that received the trimmed piece ***T***, and name the receiver of the remaining piece ***NT***.
- Agent 1 receives the remaining piece of cake 1.

Cutting cake 2

- Agent ***NT*** devides cake 2 into three pieces of equal value.
- All agents select one piece from cake 2 in the order ***T, 1, NT.***

Is everone happy? (according to slide 13)

cake 1

Agent 3 is the first to choose from cake 1 -> he is fine

Agent 2 receives one of the pieces that he himself values as equal -> fine

Agent 1 receives an untrimmed piece which he has evaluated himself as equal -> he is fine

cake 2

T chooses first -> he is fine

NT has evaluated the pieces as equal -> he is fine

Agent 1 – see above -> he is fine

Remarks

- Cake-cutting for any n: possible to have allocations that satisfy proportionality - but not envy-free
- Envy-free allocations: much harder, running time of algorithm is unbounded
- More practical: resort to optimizing a social welfare function and optimize aggregate welfare

3. Matching

Assume that agents have preferences when matching sets -

which means they themselves are matched with respect to

- jobs, goods, services, healthcare, education

What types of matching mechanisms do we have?

1. Markets (decentralized matching) – e.g. buyers and sellers
   - Good for large systems, as long as the market solution is acceptable (from a social point of view)
2. Social planer (centralized matching) – requires automated allocation process that (hopefully) can take individual preferences into consideration

Two-sided preferences

This is often referred to as the **hospitals**/**residents** problem (HR).

- We have a set of residents $r_1..r_n$
- And a set of hospitals $h_1.. h_m$
- With a certain capacity $c_1.. c_m$
- Only certain hospitals can treat certain residents, hence there is a set of relations $E \in RxH$ with acceptable hospital-resident pairs.
- Residents have preferences about the hospital they want to go to $A(r_i)$.
- Hospitals have preferences about which of the residents they prefer to treat, $A(h_j)$.

Example

I am not using subscripts here for legibility: h1=h_1 etc.

Residents' preferences			
r1	h1	h2	
r2	h1	h2	h3
r3	h2	h1	h3
r4	h2	h1	

Hospitals' preferenes and capacity					
h1	(1)	r3	r2	r1	r4
h2	(2)	r2	r3	r1	r4
h3	(1)	r2	r3		

capacity

How do we find a matching *M*, such that residents are treated in acceptable hospitals, while acounting for their preferences?

Try: Hospitals pick highest preference

M={(r3,h1) (r2,h2) } … can lead to *r1, r4* unassigned (depends on ordering)

A better approach: Gale-Shapley algorithm

1. Every resident applies to the most prefered hospital. Each hospital h_j puts c_j applicants on a waiting list in the order of hospitals preferences – others are rejected.
2. In each following step *k*, remaining residents apply to their next best choice. Each hospital h_j puts c_j applicants on updated waiting lists in order of hospital preferences - by merging new applicants with those on the waiting list from step *k-1* – others are rejected.

Application to the example

Residents' preferences			
r1	h1	h2	
r2	h1	h2	h3
r3	h2	h1	h3
r4	h2	h1	

Hospitals' preferenes and capacity					
h1	(1)	r3	r2	r1	r4
h2	(2)	r2	r3	r1	r4
h3	(1)	r2	r3		

1. *r1, r2* apply to *h1*. *r3, r4* apply to *h2*. *h2* accepts *r3, r4*. *h1* accepts *r2*. *r1* is rejected
2. *r1* applies to *h2*. *h2* accepts *r1, r3*. *r4* is rejected.
3. *r4* applies to *h1* and is rejected.

Result: ***M = {(r1,h2) (r2,h1) (r3h2)}***

(*r4* has applied to all hospitals in his preference list but cannot be matched.)

Remarks

- The resulting matching is **stable**, because every resident is assigned to the best hospital that he/she could achieve.
- The result shown here is the "resident-optimal" solution. A "hospital-optimal" solution can be contructed when we change the algorithm to hospitals offering posts to residents.
- Strategy: It can be shown that agents might have an incentive not to reveal their preferences, i.e., they might truncate their preference list after their best stable choice to influence the matching. Such kind of **strategic behavior** can also be found in other SC problems, including the previously discussed examples on voting and allocation and is an important aspect in mechanism design.

4. An example on social interaction and path dependency

Let us consider a problem with sequential choice between two alternatives, **A** and **B** (the adoption of a new technology for a product)

Payoffs for individuals are composed of

- A **private** component
- An **interaction dependent social component:** the value depends on the number of other users of the same technology

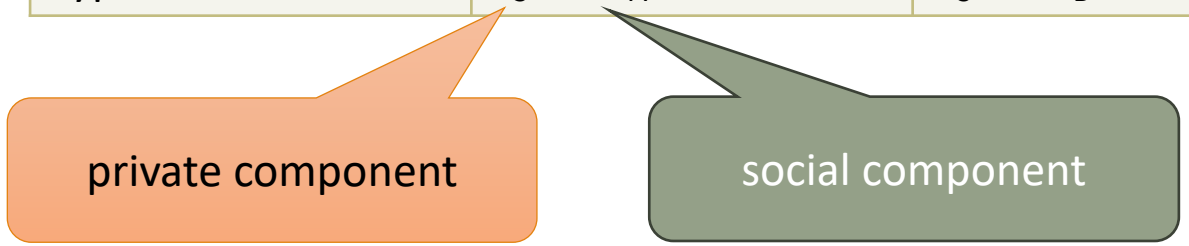
Arthur, B., "Competing Technologies, Increasing Returns, and Lock-In by Historical Events" Economic Journal 99, 1989, 116-131

even more details in:
Arthur, B., Y. Ermoliev and Y. Kaniovski, "Path-Dependent Processes and the Emergence of Macro Structure", European Journal of Operational Research 30, 1987, 294-303

Example

We have two types of agents (firms), **R** and **S**. Assume for now that the social components are identical, i.e. $r=s$. The private components differ, assume $a_R > b_R$, $a_S < b_S$. The payoffs then are given by

	Technology A	**Technology B**
Type R	$a_R + r\, n_A$	$b_R + r\, n_B$
Type S	$a_S + s\, n_A$	$b_S + s\, n_B$



Note that in the absence of the social term, agent ***R*** has a preference for choosing **A**. ***S*** would choose **B**.

Sequential choice process

- ***R*** and ***S*** agents show up in random order
- The first agents will choose according to her private component
- From the second agent the social component (externalities/previous choices) will influence the decisions

- Let us start by looking at the decission process of an ***R*** agent…

Agent ***R*** compares the following, he would choose technology **B** if

$$b_R + rn_B > a_R + rn_A$$

$$n_B - n_A > \frac{1}{r}(a_R - b_R)$$

Define $d_n = n_B - n_A$ and $n = n_A + n_B$

We can express the **market share** of technology **B** as

$$x_B = \frac{1}{2} + \frac{d_n}{2n}$$

$$d_n = n(2x_B - 1)$$

Inserting above gives

$$n(2x_B - 1) > \frac{1}{r}(a_R - b_R)$$

$$x_B > \frac{1}{2} + \frac{1}{r}\left(\frac{a_R - b_R}{2n}\right)$$

Condition for ***R***-agent to choose **B**

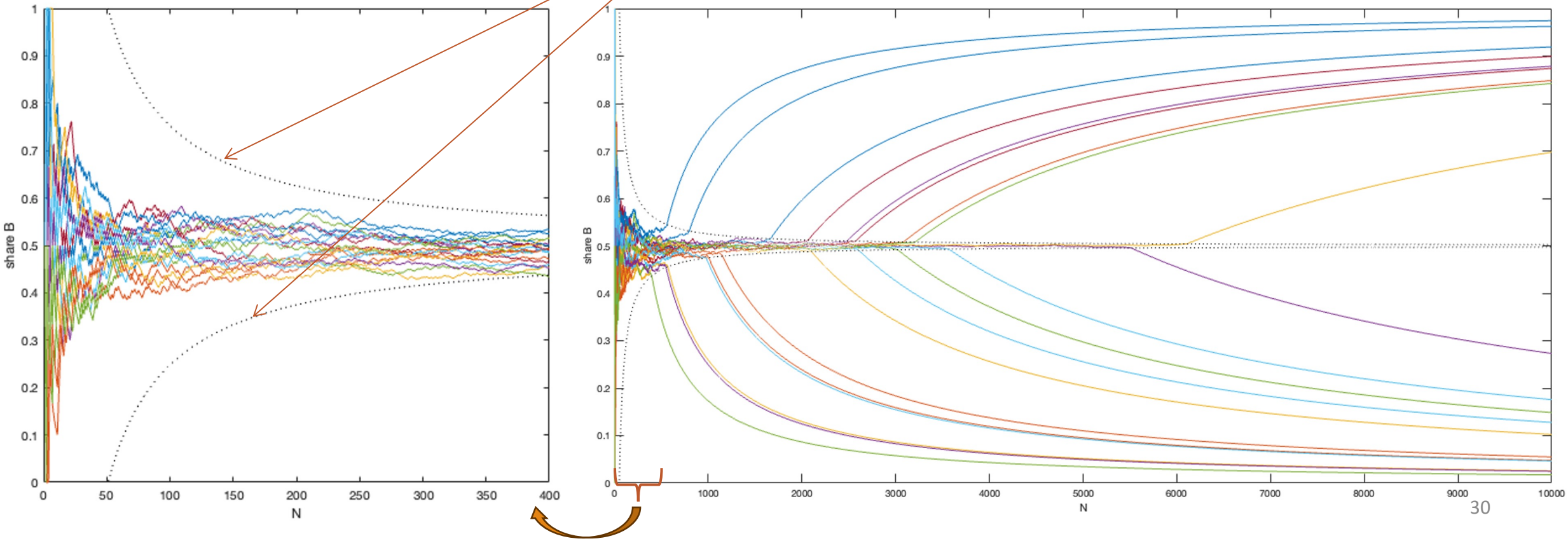
(a similar condition can be derived about when ***S*** chooses **A**)

Simulation results

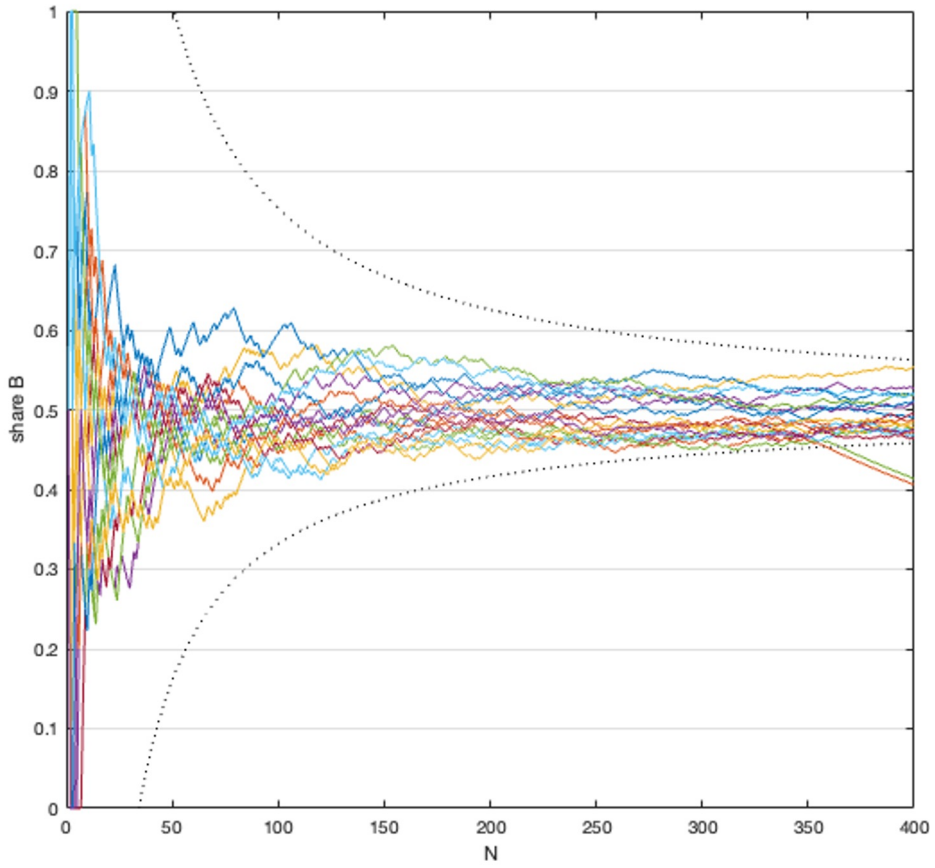
20 simulation runs with $a_R=2$, $b_S=2$, $a_S=1$, $b_R=1$, $r=0.02$, $s=0.02$, $N=10000$

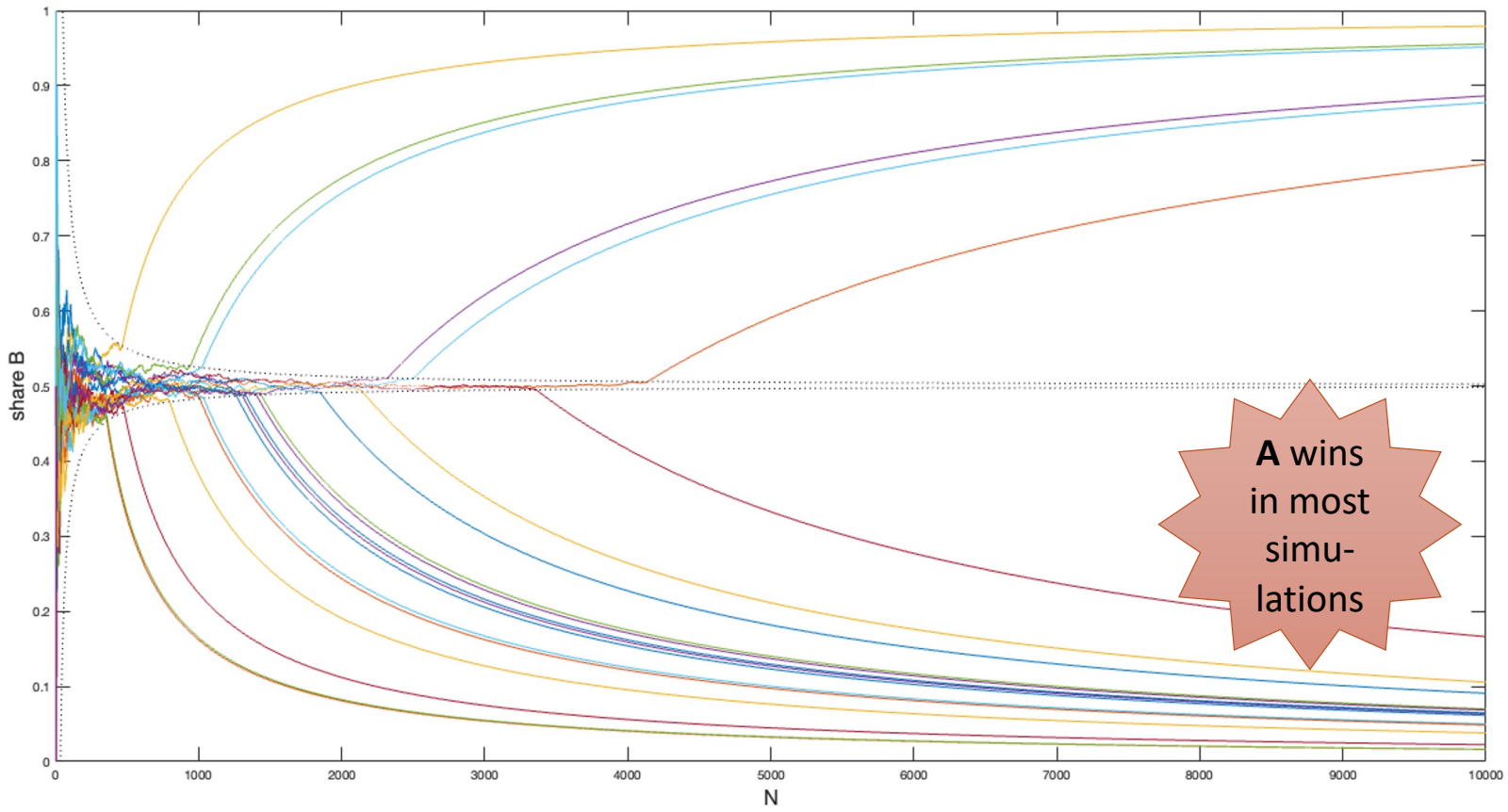
Once the conditions for ***R, S*** to choose the alternative technology are met, all new agents will choose the same alternative: **lock-in** happens!

The conditions therefore are the absorbing boundaries of a stochastic process.



An example with higher social component for **S**…
a_R=2, b_S=2, a_S=1, b_R=1, r=0.02, s=0.03, N=10000
This makes technology **A** the preferred technology. However, this does not guarantee that it will always win!





Remarks

- This process is the generalized version of the **Polya** urn process: We are drawing balls from an urn which content changes depending on past moves. Eventually we are left with an urn that contains only one kind of balls.
- The process in **non-reversible** once the absorbing boundaries are hit.
- The process is **non-ergodic**. Different samples have different distributions.
- **Lock-in** into an inferior technology is possible.
- Possible **intervention**: Push towards preferred technology as long as lock-in not reached (problem: “preferred” technology might not be obvious).

Exercise 1

Solve the HR problem below with paper and pencil!

Residents' preferences			
r1	h2	h3	
r2	h2	h3	
r3	h2	h3	h1
r4	h3	h1	h2
r5	h3	h2	h1

Hospitals' preferenes and capacity					
h1	(2)	r3	r2	r4	r1
h2	(2)	r5	r3	r2	r1
h3	(1)	r5	r4	r1	r2

We will ask for the solution in KnowledgeCheckR.

Exercise 2

For the social interaction (technology adoption) model presented in (4.)

- Derive the condition for agent ***S*** to choose technology **A**.
- Program and simulate the model, explore the parameter space.
- *Bonus question*: What would change if we allow for decreasing social returns, i.e. $r,s<0$?

No submission necessary. You can visit me in my office if you need advice (appointments by email).